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Review

A review on forming technologies of fibre metal laminates



Zerong Ding [a], Hongyan Wang [a], Jiaming Luo [b], Nan Li [a, *]

[a] Dyson School of Design Engineering, Imperial College London, Exhibition Road, London, SW7 2DB, UK
[b] Shougang Research Institute of Technology, Shougang Group, Beijing 100041, China

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ABSTRACT

Fibre metal laminates (FMLs), as a class of hybrid material taking advantages of both metals and composites, have shown great promise as lightweight structural materials in the transportation industry. Accordingly, manufacturing technologies of FMLs are attracting increasing research interests. This review emphasises the developing technologies of forming FML components, with other aspects related to FML materials being briefly introduced. First, we provide an overall review of the historical background and recent developments of FMLs, their classifications, sheet fabrication processes, and their advantages and disadvantages. Then, various forming technologies are introduced in detail, with a particular focus on stamp forming, which is considered to be the most promising approach for the high-volume production of complex-shaped FML components. Furthermore, the deformation modes and defects in forming FMLs are analysed and the challenges encountered in the existing research are thoroughly discussed. Finally, studies on modelling and process simulation of forming FMLs are reviewed and discussed. Based on the comprehensive appraisal of various aspects, current research progress and challenges related to FMLs and their forming technologies are summarised and an outlook of further developments is discussed.
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1. Introduction

Increasing concern about the climate change [1] has driven the adoption of more rigorous environmental protection policies globally, with the aim of reducing carbon footprint [2–4]. An effective solution to this task in the transportation industry is to improve the energy efficiency of vehicles by reducing their structural weight using lightweight materials. Polymer matrix composites (PMCs), especially continues fibre-reinforced polymers (FRPs) [5], and modern metallic alloys, such as aluminium alloys and high-strength steels [6,7], have been widely used for aircraft and automotive structures. However, the applications of these materials are limited by some of their disadvantages, like poor impact resistance of FRPs [8,9] and concern about fatigue failure of monolithic metal structures [10]. To overcome their respective disadvantages, the concept of hybrid material system has been proposed.

Fibre metal laminates (FMLs) are hybrid material systems interlacing metal and FRP sheets, as illustrated in Fig. 1a. They were initially developed in the aviation industry to improve fatigue resistance [11]. With other advantages, especially the high strength-to-weight ratio, FMLs have been considered to be the ideal lightweight structural material [8]. The most famous FML is GLARE, which has been successfully applied on the Airbus A380 and provided significant weight reduction with extraordinary fatigue and damage resistance performance [12]. Recently, increasing research interest has been drawn to the development of manufacturing technologies for complex-shaped FML components to enable wider industrial applications. In the automotive industry, a floor assembly enabled by the development of FML stamp forming technology, demonstrated in Ref. [13], has shown great potentials of FMLs in providing comparable structural performance with weight reduction by around 25% compared to a full-metal lightweight structure.

The substantial potential of FMLs in industrial applications has been drawing increasing research interest toward various aspects of FMLs which has been reviewed from time to time. Vlot [11] presented the history of FMLs from its birth to the development and application of ARALL and GLARE. Sadighi et al. [14] provided an experimental review on the impact resistance performance of FMLs. Morinière et al. [15] further reviewed the modelling

* Corresponding author.
E-mail address: n.li09@imperial.ac.uk (N. Li).
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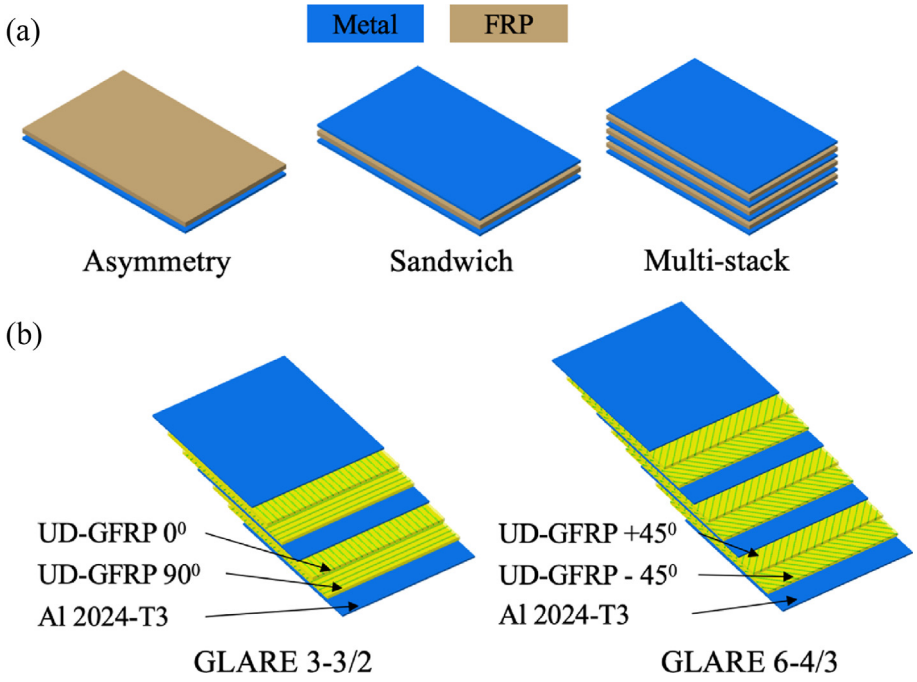


Fig. 1. Schematic illustration of (a) various FML structures; (b) example multi-stack FMLs.

development of impact damage and dynamics of FMLs. Chai and Manikandan [16] provided a review which addressed the experimental, numerical and analytical work, particularly on the low velocity impact response of FMLs. Chandrasekar et al. [17] reviewed the mechanical properties of FMLs and various failure modes under different loading conditions, as well as their hygrothermal behaviours. Sinmazçelik [8] provided a review of FMLs in various aspects, including a comprehensive background introduction, bonding process, and test methods. Existing reviews as mentioned mainly focused on the aspects of the development history of FMLs and their performance as structural materials, whereas the manufacturing technologies of FML components has rarely been reviewed. Sinke [18,19] provided an introduction to the principles of manufacturing FML components, with a particular focus on lay-up techniques and press-brake bending. Components produced using these technologies, such as large fuselage panels and stringers respectively, have relatively simple geometries and have already been commercialised. The latest developments in forming complex-shaped FML components, however, have not been thoroughly reviewed.

This paper aims to provide a comprehensive review of the developing forming technologies of FMLs. An overall introduction to the important aspects of FML materials is briefly presented first, followed by an extensive review of FML forming technologies, with a focus on die forming. In particular, stamping technologies are discussed in detail on the aspects of processing strategies and forming windows. Then, the deformation modes and challenges in FML forming are discussed, followed by a brief introduction to the modelling and numerical simulation of the FML forming processes.

2. Fibre metal laminates: development, classification, fabrication and characteristics

Although FMLs have been reviewed in various aspects as mentioned earlier, a brief introduction to FMLs is necessary to provide readers with the most up-to-date information on important aspects of FMLs, which is the prerequisite information for the review on forming technologies for FML components described in the following sections. In this section, the history of the FMLs is presented first, including more recent developments since the 2000s, with a particular focus on the development progress of FML forming technologies. The classification of FMLs is then discussed, starting with a discussion on the material constituents of FMLs, and followed by an introduction to the fabrication process of FML sheets. The advantages and disadvantages of FMLs are then briefly discussed at the end of this section.

2.1. A brief history of the development of FMLs

Since their earliest known application in the 1970s [11], FMLs have been attracting increasing industrial interests, and various types of FMLs have been developed and investigated. The history of this development is illustrated chronologically in Fig. 2.

In the 1950s, several tragical accidents caused by metal fatigue failure [10] drove the aviation industry to develop new structures and advanced materials for better fatigue resistance. Emerging from the fundamental technologies of metal bonding [20,21] and FRPs [22], Fokker, as a contractor of NASA, embedded fibres into the adhesive between titanium layers in the 1970s [11]. Such a material structure showed an improved fatigue strength, as the embedded fibres resisted the initiating of cracks and further crack growth [23]. Based on this, Aramid Aluminium Laminate (ARALL), which is an FML consisting of aluminium sheets and aramid fibre-reinforced epoxy, was developed at the Delft University of Technology and applied as wing panel materials on the Fokker F-27 in the 1980s [11,23]. A further improvement by substituting aramid fibres with glass fibres was then developed to avoid potential failure when using ARALL as fuselage materials. In the 1990s, Glass Reinforced Aluminium Laminate (GLARE) was then developed at Delft and commercialised on the Airbus A380 [11,12]. An insight into this extraordinary venture of developing ARALL and GLARE can be found in Vlot's book [11].

During the 1990s, various FMLs with different material constituents were developed and investigated. Lin et al. [24] replaced the aramid fibres in ARALL with carbon fibres, and developed

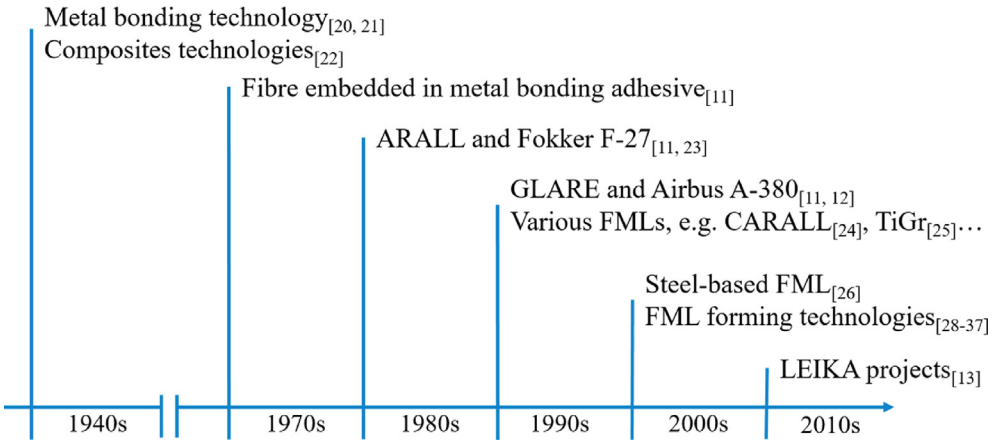


Fig. 2. Development of fibre metal laminates (FMLs).

Carbon Reinforced Aluminium Laminates (CARALL). NASA started to develop TiGr, which is an FML system consisting of titanium and graphite fibre-reinforced polymer [25]. Since the 2000s, more metals, such as steels [26] and magnesium alloys [27], have been used to construct FML structures. In addition to the exploration of various constituting materials, developing advanced manufacturing technologies for producing FML components has been a major research topic in the past two decades.

Until 2000, the major manufacturing methods used to produce FML components were bending and lay-up technique. These techniques only allowed FML components with simple geometry to be manufactured. Since the 2000s, research interest has been drawn to producing complex-shaped FML components via forming, especially die forming technologies. Mosse et al. [28] started investigating the formability of aluminium/polypropylene (Al/PP)-based FMLs. Edwardson et al. [29] investigated the feasibility of laser forming FMLs with various material constituents.

The investigation on forming FMLs drew increasing interest in the 2010s. Kalyanasundaram et al. [30] investigated the formability of Al/PP-based FMLs with double curvature dome stamping. Dau et al. [31] proposed a combined forming process, in which the steel sheet and carbon fibre-reinforced epoxy were formed and joined in one step. Behrens et al. [32] used a similar technique to stamp and join steel and carbon fibre-reinforced polyamide 6 (PA6) in one step. Mennecart et al. [33] combined deep drawing and thermoplastic-resin transfer molding (T-RTM), where steel or aluminium sheets and glass fibres were bonded, embedded and formed with a special thermoplastic matrix in one step. Sherkatghanad et al. [34] utilised hydroforming by partially replacing a solid die with pressurised fluid to form FMLs consisting of aluminium sheets and glass fibre-reinforced prepregs. Chernikov et al. [35] investigated the feasibility of forming GLARE-like FMLs with punch force generated by electro-magnetic pulse. In addition to the various die-forming technologies as mentioned above, other forming methods were also investigated during these decades. Russig et al. [36] used shot peening forming (SPF), which was an established process for fuselage parts, to form GLARE and showed a typical curvature radii of 2500 mm for fuselage panels can be achieved. Al-Obaidi et al. [37] formed Al/PA6-based FMLs with basalt fibre reinforcement with heat support incremental sheet forming. These forming technologies mentioned will be discussed in detail in section 3.

Among the development of these FML forming technologies, LEIKA was a particularly remarkable project partnering with several industrial and academic research institutes [13]. They developed advanced FML forming technologies [38] which enabled the production of a multi-material floor panel structure with 25% further weight reduction compared with a full-metal structure [13]. This three-year project demonstrated the significant lightweighting opportunities presented by FMLs and their advanced manufacturing technologies in the automotive industry.

2.2. Classification of FMLs

FMLs can be classified either by their structural arrangements or material constituents. There are variants of FML structural arrangements, as illustrated in Fig. 1a, from asymmetric to multi-stacking laminates, based on various performance requirements and manufacturing considerations [39–41]. Especially for multi-stacking laminates, specialised performance features can be achieved by alternating the stacking arrangement and the fibre orientations. For example, GLARE can exhibit different features by alternating the orientations of the unidirectional fibres embedded in the polymer. As shown in Fig. 1b, GLARE 3 was designed for optimal fatigue resistance; whilst GLARE 6 was designed for enhancing the shear and off-axis properties [42].

Alternatively, a more recognised classification for FMLs is based on their material constituents. The constituting materials, in theory, can be selected freely. However, there are many restrictions to be considered to construct reliable FMLs, to avoid extreme internal stress, galvanic corrosion, voids and volatiles, etc. Availability and cost are also important factors to be considered when selecting the material constituents of FMLs [43]. FML material constituents in the literature are summarised and presented in Fig. 3. Based on the material constituents, FMLs can be primarily classified by the base metals and then the composites, which each consists of matrix and fibre reinforcement. Compared with the classification in previous reviews [8,16,17], this figure includes the recent exploration of steel-based FMLs and thermoplastic-based FRPs.

Aluminium-based FMLs are the most commonly investigated and developed FML as aluminium is the most widespread structural material in the modern aviation industry. Several aluminium-based FMLs, such as ARALL and GLARE, have been commercialised as fuselage and wing panel materials, as mentioned in the previous sections [11]. Titanium-based FMLs have also been investigated by leading aerospace research institutes such as NASA [25,53]. Recently, studies have commenced on steel-based and magnesium-based FMLs as potential lightweighting solutions for the automotive industry [13].

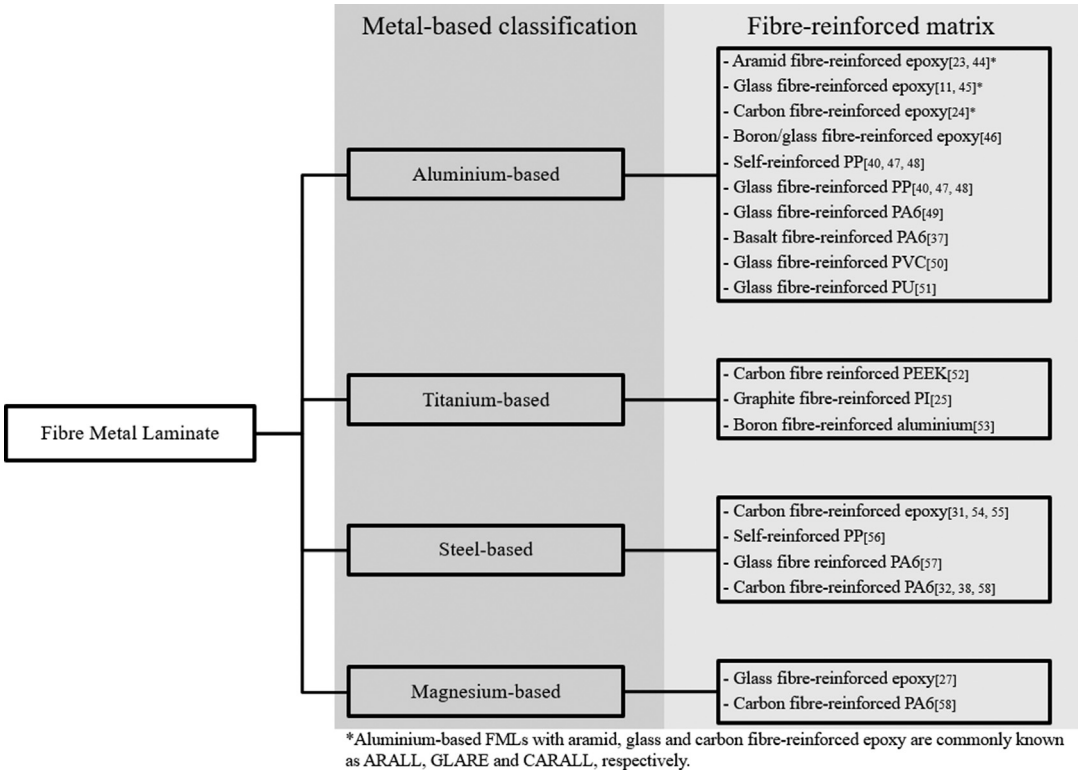


Fig. 3. Classification of FMLs based on material constituents [44,46,52].

The most commonly used interlacing composite layers in FMLs are FRPs, which can be further classified by their reinforcing fibres and polymer matrices. It can be seen from Fig. 3 that the most commonly investigated and used fibres in FMLs are carbon fibres and glass fibres. Nevertheless, when designing FMLs with carbon fibres, the risk of galvanic corrosion between the fibres and the metals need to be considered [59]. In addition, the reinforcement structure can be either unidirectional (UD) or woven as shown in Fig. 4.

There are various types of matrix materials in existing FMLs. Thermoset polymer, such as epoxy, is a relatively well-developed matrix solution to FMLs including GLAREs [45]. However, the long curing time of such matrix increases the production cycle time and consequently limits the production volume [8]. Thermoplastic polymer has been growing rapidly recently due to various potential advantages including better recyclability, and superior toughness and chemical resistance in service, as well as their rapid processing time [60]. When constructing FML components, the process cycle time can be reduced from 4 min [31] to seconds [32] by using thermoplastic matrices. Therefore, thermoplastic matrices are becoming popular in constructing and forming FMLs in recent studies. In addition to polymers, metals can also be used as the matrix materials to construct metal matrix composites (MMC), as well as FMLs. For example, NASA constructed FMLs by bonding boron fibre-reinforced aluminium with titanium sheet covers [53].

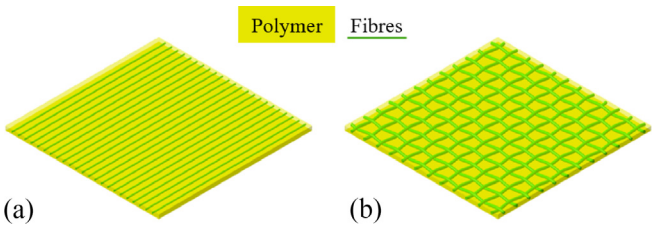


Fig. 4. Schematic illustration of typical fibre reinforced arrangements: (a) unidirectional (UD); (b) woven.

2.3. Fabrication of FML sheets

Fabricating FML sheets is similar to the process of fabricating FRP sheets [61] but varies in their pre-treatments and post-treatments. A typical FML fabrication process is described as follows:

2.3.1. Preparation: metal surface treatment

Similar to metal bonding [62], the surface of the metal sheets need be prepared before FML fabrication. Metal surface treatments generally fall into three categories, which are chemical/electro-chemical treatments, mechanical treatments, and applying adhesion prompter [8]. Different methods are recommended by the ASTM International standards D2561-01 [63] for a wide variety of metal alloys, as well as for the compatibilities of the adhesives to be used. Typical surface treatments employed for fabricating FML sheets and producing FML components are listed in Table 1. It should be noted that surface treatments are normally performed

Table 1
Examples of key surface treatment in FMLs production.

Metal	Matrix	Metal surface treatment	Ref.
Aluminium	Epoxy	Anodizing	[18]
Aluminium	Epoxy	FPL etching	[64]
Aluminium	PP	5% NaOH solution	[40]
Titanium	PI	Pasa-Jell 107	[25]
Steel	PA6	Galvannealing	[57]
Steel	PA6	Adhesion promptor	[13]

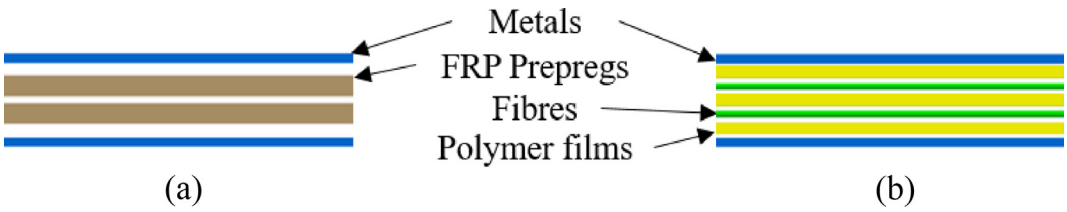


Fig. 5. Stacking strategies for FML fabrication: stacking metal layers with (a) FRP prepregs; (b) alternate fibre and polymer film layers.

with several distinct steps sequentially, and only the featured steps are included in table. A proper surface quality is crucial for optimal bonding quality. In the review of Sinmazçelik et al. [8], various surface treatments were introduced in further detail.

2.3.2. Stacking and bonding

The prepared metal sheets and FRPs are then stacked and bonded with heat and pressure applied. This process is similar to production of FRP sheets [61] and different stacking strategies are used, based on the choice of raw materials. A typical practice is to stack FRP prepregs with metal sheets, as shown in Fig. 5a [40]. Instead of stacking FRP prepregs, teams from the LEIKA project used an alternative approach named the "film stacking process" [13]. In this process, alternating layers of thermoplastic films and carbon fibre fabrics, were stacked with metal layers, as shown in Fig. 5b. The FML stack is then bonded in an autoclave [45] or a hot press [40].

2.3.3. Post-treatment: stretching

Due to the residual stress introduced by the different thermal expansion of metal and FRP layers in FMLs, the material's fatigue performance could be decreased. This condition can be improved by certain post-treatments. For example, when fabricating ARALL, a post-stretch process was developed to eliminate or reduce the residual stress by elongating the cured laminates until the aluminium sheets yield [45,65].

2.4. Advantages and disadvantages of FMLs

—Advantages

Combining the merits of both metals and FRPs, FMLs have superior advantages compared to either monolithic metal or FRPs. The most significant advantage of FMLs is their fatigue resistance. As the initial design objective, the fatigue crack growth rate of GLARE is 10–100 times slower than monolithic aluminium alloys [66]. FMLs also exhibit other excellent mechanical properties, such as a high strength-to-weight ratio and high impact resistance. An example of these properties in comparison with monolithic aluminium sheets is provided in Table 2.

Benefiting from the interlacing structures, FMLs have other advantages such as moisture, corrosion, and flame resistance. Moisture absorption is a common issue in FRPs. It is slowed in FMLs due to the metal layers acting as barriers [69]. Reciprocally, the barrier role played by the FRP layers preventing the through-thickness corrosion [70]. Furthermore, the glass fibres in GLARE increase their fire resistance with its high melting point (T_m) [70]. For further details, it is recommended to refer to Sinmazçelik et al.'s review [8], where the advantages of FMLs were comprehensively discussed.

Table 2
Benchmark of mechanical properties between a monolithic Al sheet and GLARE 3 [66–68].

	Aluminium 2024-T3	GLARE 3-3/2
Ultimate tensile strength (MPa)	455	717
Density (kg/m^3)	2780	2520
Specific strength (kN·m/kg)	163.7	284.5
Low velocity impact failure energy (J)	9	>37
High velocity impact failure energy (J)	37	95

— Disadvantages

As hybrid material systems, the fact of inhomogeneous through-thickness properties of FMLs increases the difficulties in quality inspection during production and in-service maintenance [71]. Another major disadvantage of FMLs is the challenges in manufacturability, despite the progress of FML forming technologies being made since the 2000s. Although the processing time can be potentially shortened by the application of thermoplastic matrices, there are still unsolved problems in forming defect-free FML components with complex shapes at large production volumes.

3. Forming technologies for FML components

FML components can be manufactured by forming metal panels and fabricating FRP panels into designed shapes separately and then bonding them with adhesives [31]. However, this method requires a long cycle time and multiple tool sets, and is limited by the achievable as-formed geometric accuracy of the individual sub-components. Direct forming FMLs has therefore been attracting more interest. In this section, representative forming technologies for producing FML components are discussed.

3.1. Press brake bending

As a well-developed metal working process, press brake bending can also be used to produce single curvature FML components such as stringers [19], as shown in Fig. 6. The design of such components is, however, limited by the low failure strain of the embedded fibres. Thin FMLs (thickness less than 2 mm) with UD reinforcement can be bent easily as monolithic metal sheets when the bending line is parallel to the direction of the embedded fibres, and the minimum bending radius and springback is dominated by the metal covers, and similar to the monolithic metal alloys. Larger springback will occur when the bending line is perpendicular to the fibre direction, and the minimum bending radius will rapidly increase when the thickness is larger than 2 mm. In addition, bending thicker FMLs will increase the risk of delamination due to the increased interlaminar shear stress [19,72].

3.2. Lay-up technique

Large components such as fuselage and wing panels can be manufactured by a lay-up technique, as shown in Fig. 7. Similar to the manufacturing of FRP components, each layer is stacked onto a mould with a designated sequence. With a large area-to-thickness ratio, the metal sheets can easily deform with FRP prepregs into the mould without significant forming force. The stacked panel is then vacuum bagged and sent to an autoclave for curing [18]. The

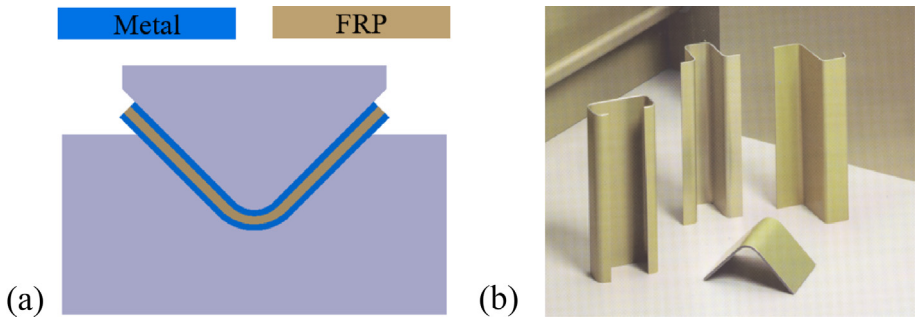


Fig. 6. Press brake bending of FMLs: (a) schematic illustration; (b) example of FML components produced by press brake bending [18].

applicable geometries of this technique are significantly restricted by the curvature, and only single or shallow double curvature panels with relatively large radii can be formed [18,19].

3.3. Shot peening forming (SPF) and incremental sheet forming (ISF)

Shot peening forming (SPF) and incremental sheet forming (ISF) utilise indentation impact to create local plastic strain, as shown in Fig. 8, gradually leading to a macroscopic curvature. In SPF, the indentation impact is introduced by shooting round peens; whilst in ISF, it is provided by a round tipped tool.

SPF is a developed technology in the aviation industry [73,74]. It can be used to produce large panel components such as fuselages. Russig et al. [36] investigated the feasibility of implementing SPF to form GLARE panels. It was evident in their research that convex geometries could be formed, and the radii could be as small as 1000 mm for GLARE 3, which made it a viable technique to produce A380 fuselage panels. They also pointed out that non-destructive inspection was necessary for parts produced by SPF and the Lock-In thermography technique was recommended.

ISF is a rapid prototyping and small-quantity production technology for both metal and polymer sheets [75]. Jackson et al. [76] reported that ISF is viable for forming various sandwich panel including Hylite, which comprises aluminium covers and PP core. Al-Obaidi et al. [37] introduced a hot air pump to heat the FML blank during ISF. A cone-shaped component consisting of aluminium alloy covers and basalt fibre-reinforced PA6 core was formed. The feasibility of forming complex FML components with ISF was evident. However, due to the relatively long cycle time, the production scale was limited.

3.4. Laser forming

Laser forming, as shown in Fig. 9, has been used as a technique for rapid prototyping, adjusting, and aligning for metallic components. In this process, a surface of the component is heated by a laser beam irradiation and the resultant thermal stress leads to a change of shape at the beam scanning path [77]. Edwardson et al. [29] investigated the feasibility of adopting this technology to form various types of FMLs, including the GLARE. In their study, multiple scans were conducted to achieve a large bending angle, and a part-cylinder was formed. Carey, in his thesis [78], further investigated the effects of stacking sequences and fibre orientation of the FMLs on laser forming, as well as the effects of various processing parameters, such as laser power, number of passes, etc. Though it was indicated that the scan path could be customised to make it possible for double curvature forming, the current studies are still limited to forming single curvature components. Also, the residual stress introduced by the unheated lower layer, which is the obverse surface not irradiated

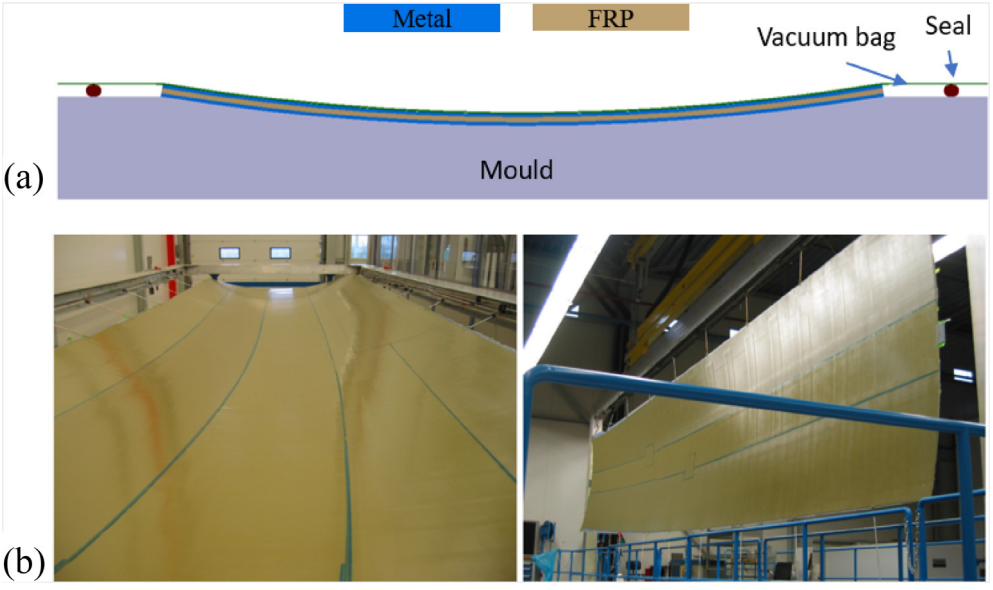


Fig. 7. Lay-up techniques: (a) schematic illustration, (b) fuselage panels produced through lay-up processes [18].

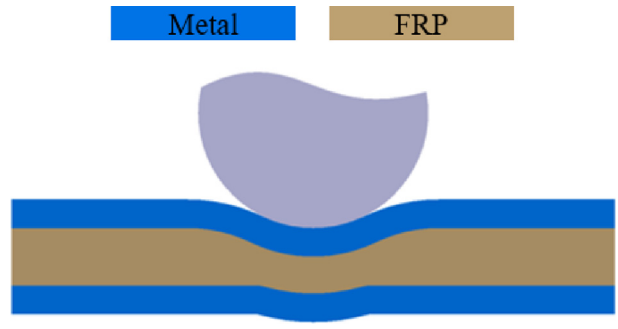


Fig. 8. Schematic illustration of indentation impact during SPF or ISF of FMLs.

by laser beam, is a significant factor causing forming defects, such as cracks and delamination, and in-service risks.

3.5. Die forming

In die forming, material blanks are pressed against a die mould and shaped under the deforming force introduced. Die forming, especially stamping, is a well-developed mass production technology for forming sheet metal components [79], and has also drawn extensive interests in producing FRP components [80]. Recently, die forming has also been utilised to form FMLs, mostly with aluminium alloy or steel covers (there have also been a few studies on magnesium-based FMLs [13,58]). Researches on forming FMLs with aluminium alloy covers began in the 2000s [28,30,33,40,47–50,81–94], while research on forming FMLs with steel covers started, relatively late, in the 2010s [31,32,38,39,55,56,58,84,95–100]. Among these studies, most research groups used thermoplastics as the matrix for shorter processing time. To improve the formability of the FRP layers, most of these studies formed FMLs at elevated temperatures. There have been three variants of die forming techniques utilised to form FMLs, as shown in Fig. 10.

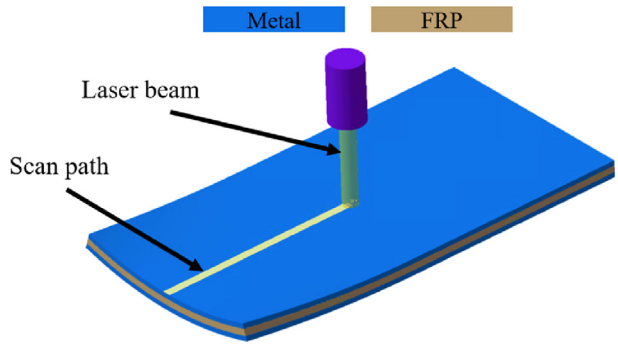


Fig. 9. Schematic illustration of laser forming of FMLs.

– Stamping

Stamping is the most widely used die forming method. The tool sets typically comprise a set of matched dies, and a blank holder. Tool sets are usually heated for stamping process of FMLs. Mosse et al. [48] set the tool at either room temperature or an elevated temperature, so that the preheated FML blank could be cooled at a desired cooling rate. In the work of Behrens et al. [32], the blank was heated on the tool to shorten the process chain. Dau et al. [31] pointed out that a heated tool can enable a cycle time as short as 4 min to form epoxy-based FMLs. Additionally, Nam et al. [56] used an open die during forming to capture the surface strain evolution with an optical system during forming.

– Hydroforming

Hydroforming, which utilise pressurised fluid in the forming process, is a widely used metal forming technology, especially for tubular components. For sheet components, die forming can be adopted by replacing a portion of the lower die with pressurised fluid, against which the sheet is deformed by the upper die. Although hydroforming is less commonly used because of its higher cost compared to conventional stamping, it can nevertheless be attractive for its higher formability and superior surface quality [101]. Recently, this technology has been used to form FMLs [34,51].

– Electro-magnetic forming

Electro-magnetic forming is a high speed, non-contact forming technology for tubular and sheet components. It uses a pulsed magnetic field to apply force onto the workpiece [102]. Recently, Chernikov et al. [35] investigated the feasibility of using this method to form FML components.

3.6. Summary

A feature comparison of the FML forming technologies discussed in this section is summarised in Table 3. Among these technologies, press brake bending and lay-up of FML components are relatively well-developed and have been commercialised to produce structural stringers and large aircraft skin panels, respectively. The feasibility of forming FML panels through laser forming and SPF has also been investigated. However, the formability of these forming technologies is inadequate to be applied towards producing FML components with complex geometries. To form complex-shaped components, ISF and die forming have been investigated. ISF is able to form complex-shaped FML components,

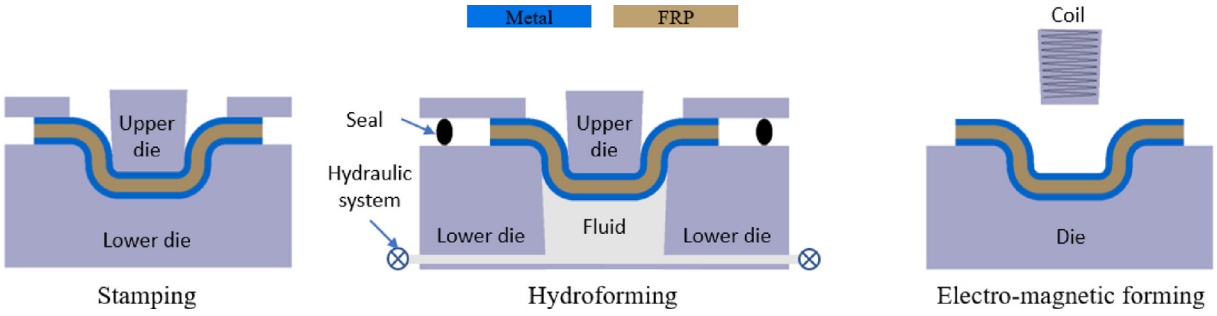


Fig. 10. Variants of die forming methods.

Table 3
Feature comparison of different FML forming technologies.

	Shape	Radii	Cycle time	Production volume
Bending (Press brake)	Single curvature	Small	Short	Large
Lay-up	Single, shallow double curvature	Large	Long	Medium
Laser forming	Single curvature	Large	Long	Small
Shot peening forming (SPF)	Single, shallow double curvature	Large	Long	Medium
Incremental sheet forming (ISF)	Complex double curvature	Small	Long	Small
Die forming/Stamping	Complex double curvature	Small	Medium	Large

but is not suitable for large volume production due to its long cycle time. Die forming, especially stamping, as a traditional mass production technology, has been showing promise and attracting increasing research interests in forming FMLs. Therefore, stamping for FMLs is further discussed in detail in the following sections.

4. Stamping for FML components

Similar to the hot stamping process for sheet metals, which has been utilised to produce structural elements on automobiles such as roof rails and B-pillars [103], most FML blanks are formed at elevated temperatures. However, due to the multi-constituent material system, the preparation steps before forming FML blanks are more complicated than forming monolithic metal sheets and the factors to determine key forming parameters, such as the temperature windows, are different. In this section, different processing strategies for forming FMLs are presented, followed by a discussion on the forming temperature windows of FMLs consisting of various materials.

4.1. Different processing strategies

Fig. 11 schematically illustrates four typical processing strategies for forming FMLs, involving various raw materials and pre-process arrangements in terms of stacking and heating, before forming.

– Heating bonded FML blanks

In this process, fabricated FML sheets are directly heated and then formed. Mosse et al. [28,40,47,48,94] formed Al/PP-based FMLs after preheating the fabricated sheets. Nam et al. [56,95,96] and Rahiminejad et al. [98] extended this processing strategy to steel/PP-based FMLs.

– Heating semi-finished FML stacks

In this process, metal sheets and FRP sheets (laminated or loose prepregs), separately sourced, are stacked and then heated together, prior to forming. In the study of Behrens et al. [49], the aluminium sheets and glass fibre-reinforced PA6 sheets were only stacked but not bonded before heating; then, the stack was heated and subsequently bonded in the heated, closed dies during the forming process. Hahn et al. [104] heated semi-finished FMLs stacks, provided by the LEIKA project partner, in which the FRP core and metal covers were bonded by adhesion agent. Behrens et al. [32] further optimised this process by replacing the FRP laminate with unconsolidated FRP prepregs. Also, they placed additional PA6 films between the steel covers and the carbon fibre-reinforced PA6 core for better bonding quality. Lee et al. [39] also heated semi-finished FML stacks consisting of steel and carbon fibre-reinforced epoxy prepregs before forming.

– Separate heating

In this process, metal sheets and FRP sheets, separately sourced, are separately heated to different temperatures and subsequently stacked, prior to forming. Guo et al. [55] heated uncured epoxy-based CFRP sheets and steel sheets separately to 120 °C (the melting point of the polymer) and 550 °C (the warm forming temperature of the steel), respectively; and then, the sheets were stacked, and subsequently formed into a box.

– Compression molding and resin transfer molding (RTM)

Compression molding and resin transfer molding (RTM) are techniques developed to produce composite components. Mennecart et al. [84] combined compression molding and stamping to form FML components. In this process, the bottom steel layer was placed on the heated die with dry glass fibres superimposed on.

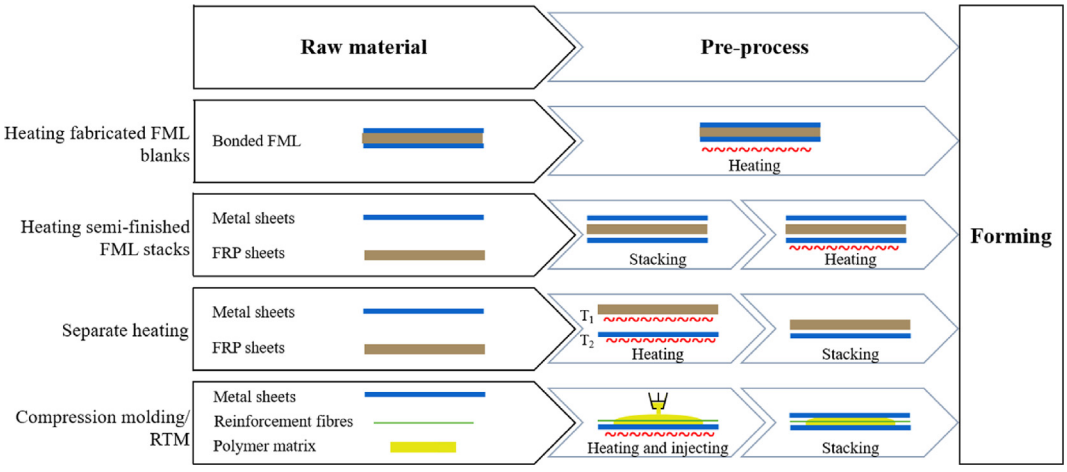


Fig. 11. Processing strategies developed in existing research.

Thermoplastic liquid was then applied onto the blank before the top steel layer being stacked, after which the stacked sheet was formed. An alternative method developed by them was to combine the T-RTM (RTM with thermoplastic matrix) and stamping, whereby the thermoplastic liquid was injected through concentric holes on the upper die and the top steel layer, which was stacked on the top of the bottom steel and fibre layers prior to transferring to the tool. The polymer can also be injected between the metal sheets through the edges of the stacked blanks. In addition, they also mentioned that the injection can take place before, during, or after forming [33,84]. A special thermoplastic liquid, named Elium, was used in this process [84], but other polymers, such as cast-PA6 (PA6G) and polymethyl-methacrylate (PMMA), were also suggested to be potential matrix solutions [33].

In the above-mentioned strategies, heating fabricated FML laminates is not viable for forming thermoset-based FMLs, because thermoset polymers cannot be reshaped after curing. The strategies of heating semi-finished FML stacks and heating separate metal and FRP sheets can shorten the overall production chain, by skipping the step of fabricating FML sheets; also, these two strategies can be adopted to form both thermoset-based and thermoplastic-based FMLs. In addition, separate heating allows the thermo-mechanical behaviours of metal constituents to be taken into consideration and offers an opportunity of improving the formability and as-formed strength of FML components. Compression molding and RTM are also viable strategies for forming both thermoset-based and thermoplastic-based FMLs, provided the polymer used has a suitable viscosity.

After forming, the part is usually dwelled in the closed die for bonding. In some studies, post-form thermal treatment was performed to increase the bonding quality and efficiency. For thermoset-based FMLs, Guo et al. [55] heated the die to the epoxy curing temperature (150°C) for curing and bonding. For forming thermoplastic-based FMLs, Behrens et al. [32] cooled the die and the components to establish the bonding between the metal covers and FRP core. As they further mentioned, geometric accuracy can also be improved, benefitting from the post-form thermal treatment. Thus, further investigation on the effect of cooling rate and removal temperature was suggested.

4.2. Forming temperature windows

The forming temperature is considered to be the most important process parameter in FML forming. It is noted that the temperature windows for FML forming in existing researches are found to be mostly ensuring the optimal formability of the FRP layers, regardless the material choice of the metal layers. Fig. 12 summarises the forming temperature windows obtained from existing research on forming FMLs or forming thermoplastic-based FRPs. The forming temperature varies depending on the material constituents and is mainly determined based on the glass transition temperature (T_g) and melting temperature (T_m) of the polymer matrix. The melt-state forming and solid-state forming represent that the forming temperature is above and below T_m of the polymer matrix, respectively.

– FMLs with PP matrix

Zampaloni et al. [106] noted that the fibre-reinforced polypropylene (PP) should be formed between T_g and T_m. With the low T_g of PP, both PP-based FRPs and PP-based FMLs can be formed at room temperature. Nevertheless, Gresham et al. [83] noted that a higher forming temperature could improve the formability of Al/PP-based FMLs, but will also increase the risk of wrinkling. Mosse et al. [48] identified that the optimal forming window for Al/PP-based FML was 120–140 °C, which was achieved by preheating the FML blank to 160 °C and transferring it to the tool which was preheated to 80 °C. Rahiminejad et al. [98] extended this method to FMLs with galvanised steel covers by taking account of the interaction between blank holding force (BHF) and temperature. The results showed that, for a higher BHF of 14 kN, a higher temperature (110–140 °C) can reduce the risk of material fracture but increase the risk of wrinkling and delamination. For lower BHFs (2 kN, 7 kN), wrinkling occurred at the room temperature and delamination occurred at elevated temperatures. Kalyanasundaram et al. [30] showed that the FMLs behaved differently when forming above or below the melting point T_m, as the laminates could have a higher level of inter-ply slip at higher temperatures whilst act as an intact thick sheet below T_m. Rajabi et al. [92] concluded that the drawing force dropped significantly with an increase in temperature from below to above T_m.

– FMLs with PA6 matrix

Behrens et al. [112] noted the general recommendation is that the forming temperature for thermoplastic-based FRPs should be 40 °C higher than their T_m, but not exceed their decomposition temperatures. They formed glass fibre-reinforced PA6 at 260 °C, since the melting point of PA6 is about 220 °C. Behrens et al. [32] then investigated stamping for steel/PA6-based FMLs. It was found

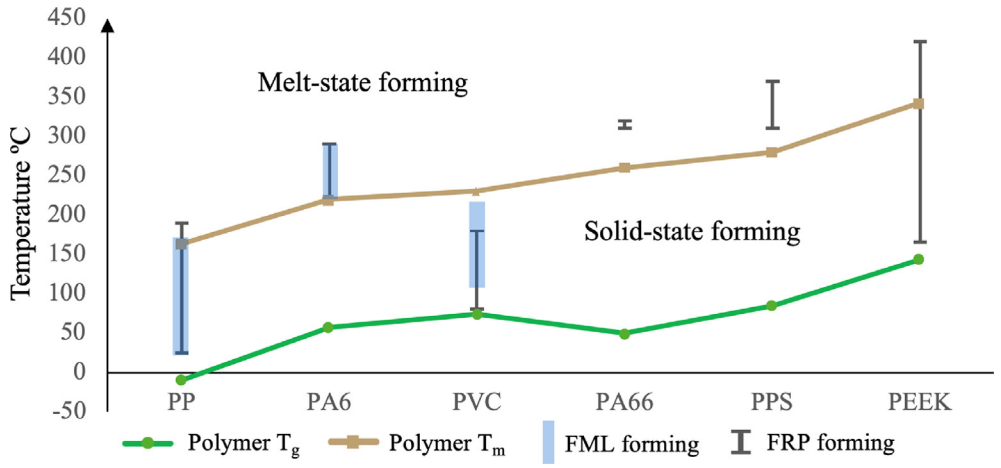


Fig. 12. Forming temperature windows of FMLs and FRPs with various thermoplastic polymer matrices (produced with data from: FML forming [28,30,32,38,40,47–50,56–58,83,85,87,91,95,98,104], FRP forming [50,105–118]).

that the temperature for optimal bonding quality was 250 °C, while excessive temperature (280 °C) would result in outgassing of the polymer and create gas cavities in the matrix. In addition, Hahn et al. [104] determined that insufficient temperature (160 °C) could not allow desirable inter-ply slip during bending, and a preheating temperature of 240 °C was used in their following experiments.

– FMLs with PVC matrix

Zal et al. [50] investigated the forming temperature for Al/polyvinyl-chloride (PVC)-based FMLs. Due to PVC's sensitivity to temperature, the temperature window was kept under 220 °C to avoid degradation. They also defined forming temperature for PVC-based FMLs should be 20 °C higher than for PVC-based FRPs. It should be noted that the melting point shown in Fig. 12 is actually the consolidation temperature (230 °C) used by Zal et al. [50]. Because there are various PVCs with different melting points, the authors did not specify the melting point of the PVC they used.

– FMLs with thermoset matrix

For stamping thermoset-based FMLs, warm forming was also considered as the matrix needs to be cured at an elevated temperature [31]. Guo et al. [55] separately heated the prepregs and metal sheets for better formability (as mentioned in section 4.1), and successfully formed a box-shaped component. Heggemann and Homberg [99] reported that local preheating improved the forming quality by better regulating the polymer flow during forming.

– Forming of other FRPs investigated

Forming temperature windows for FRPs based on other popular thermoplastic matrices, such as PA66 [113], PPS [114,115] and PEEK [116–118], have also been studied. It can be seen that all of them could be formed above the melting point. In addition, the feasibility of forming PEEK-based FRP at solid state was investigated [118]. Considering the relatively high melting temperature, solid-state forming could be an attractive and cost-effective solution for PEEK-based FRPs. However, studies of FMLs based on these polymer matrices have not been found.

It can be summarised that the temperature windows applied in FML forming were mostly determined by the polymer matrix in FMLs. Apart from the work of Guo et al. [55] as mentioned earlier, preferable thermomechanical properties and forming temperatures for metals were, however, rarely considered. The metals used in FMLs, found in existing applications and studies, are mostly suitable for cold-working, such as aluminium 5505H34 [48], aluminium AW-5754 [49], steel HC220Y [32], steel DC04 [31,54], etc.

5. Deformation modes and mechanical characterisation in forming FML components

To optimise the formability of FMLs, it is important to understand their deformation modes and mechanisms during forming. The behaviour of FMLs during forming is organically determined by the deformation mechanisms of metals and FRPs, as well as the interactions between the material constituents.

As a better developed technology, the deformation modes involved in the metal forming process is relatively well understood [119,120]. The deformation mode of FRPs varies with different arrangements of the reinforcing fibres, primarily including woven and unidirectional (UD), with UD being further classified as uniaxially oriented UD or cross-plied UD. The understanding and control of the behaviours of FRPs during forming, notwithstanding increasing research interests [80,121], remains a major challenge for forming FML components. In addition, the interfacial interactions between the metal and FRPs also need to be noted during forming FMLs. In this section, a brief introduction to these mechanisms is presented. The primary deformation modes of the material constituents in FMLs during forming are summarised and illustrated in Fig. 13.

5.1. In-plane

Sheet metal forming is commonly considered to be plane stress deformation, where the small stress in the direction perpendicular to the blank can be neglected. The material mechanical responses to this loading condition are usually characterised by uniaxial tensile tests; in addition, forming limit diagram (FLD) is a useful tool to understand the formability of sheet metals, and can be obtained from various test methods, such as Nakazima test [122,123], Marciniak test [122,124], and biaxial test [125].

The in-plane behaviour of FRPs during forming is more often referred as intra-ply deformation, as the plane in discussion is usually considered a single ply of FRPs. The dominating

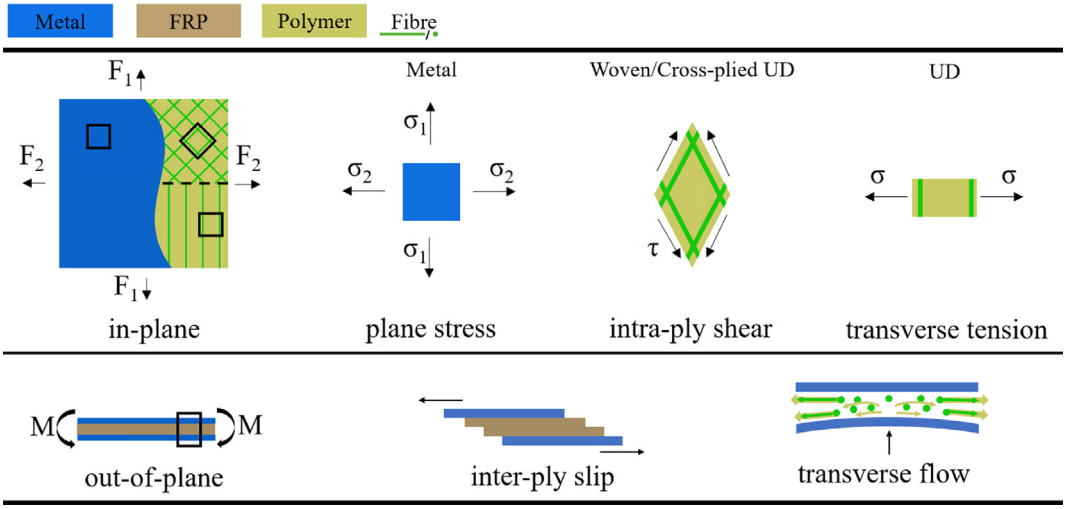


Fig. 13. Primary deformation modes in FML forming.

deformation mode of woven fabrics or cross-plied UD laminates during forming is intra-ply shear, corresponding to the reorientation of nearly-inextensible fibres [126]. Transverse tension in uniaxially oriented UD prepregs can be a dominating deformation mode during forming as the material under transverse tension is less stiff than that under shear [127]. In addition, intra-ply longitudinal tension of all FRPs is considered a less important factor due to small elongation at break of fibres.

For woven fabrics, intra-ply shear is considered to be the primary deformation mode in forming [128,129]. During forming, woven fibres rotate in a 'pin-jointed' fashion under shear load. When the locking angle is reached, where further shear and rotation is impossible, compressed fibres would experience buckling or out-of-plane deformation and eventually result in macroscopic wrinkling [130]. In addition, the rotated fibres reduce space between them [131], where local thickness increases caused by material incompressibility can be expected [132]. For uniaxially oriented UD prepregs, intra-ply shear often refers to the longitudinal shear in the plane parallel to the ply surface, despite the existence of transverse shear through the thickness [140]. Potter [127] reported that, in uniaxially oriented UD prepregs, the material under shear deformation mode was ten times stiffer than that in transverse tension at room temperature. Therefore, it is difficult to balance material flow in forming UD prepregs with uniaxially oriented plies. Instead, forming cross-plied UD sheets is more likely to be adopted. For cross-plied UD sheets, Potter [134] further noted that their deformation modes showed similarity to the woven fabric, thus could be predicted by 'pin-jointed' model which is developed for woven fabrics.

The intra-ply shear behaviour is usually characterised through bias-extension or picture frame tests for woven fabrics [128,135]. Bias-extension test has also been used to characterise the intra-ply shear behaviour for cross-plied UD sheets [134]. For uniaxially orientated UD sheets, McGuinness and Brádaigh performed picture frame tests by supporting the UD sheet specimen with vacuumed diaphragms. However, the effect of diaphragms cannot be measured to determine the accurate response of the UD sheets. Potter [127] performed off-axis test for unsupported UD plies. It is noted that when the angle between the fibre orientation and loading direction is larger than 15°, the dominating deformation mode would be transverse tension rather than intra-ply shear. In addition, Haanappel and Akkerman [136] conducted a torsional bar test to characterise the intra-ply shear of uniaxially oriented UD sheets. Furthermore, although the standardised V-notched rail shear method [137] was used to test the in-plane shear of FRPs [138], no literature has been found to use this method for characterising their behaviours during forming.

5.2. Out-of-plane

Out-of-plane bending is another important deformation characteristic in forming both sheet metals and FRPs. The bending behaviour in sheet metal forming is relatively well understood [139]. The bending behaviour in forming a FRP prepreg, however, has received little attention as their bending resistance is orders of magnitude lower than intra-ply shear resistance [128]. For forming laminated materials including FRP sheets and FMLs, inter-layer interactions and through-thickness behaviour, which primarily represented by inter-ply slip and transverse flow, need to be taken into consideration.

Inter-ply slip is an important deformation mode to accommodate the bending of laminated materials, especially during solid-state forming. By allowing relative movement between layers, inter-ply slip can reduce the wrinkling of the inner layer of a laminate [104,128,140] and cracking of the outer layer [104]. There have been extensive studies on tribology to characterise the ply-ply and tool-ply slip in forming FRPs. In particular, the results from tool-ply tests can be studied to better understand the interfacial characteristics between the metal and FRP layers in FMLs. Various test methods, including 'pull-out' and 'pull-through' tests, have been developed to characterise the interfacial behaviours in forming FRPs. Recently, a benchmark study of several different test methods has been conducted by Saches et al. [141] with PP-based FRPs. In addition, Mosse et al. [81] conducted a 'pull-out' test with a specially fabricated FML specimen to understand the shear behaviour between the bonded aluminium and glass fibre-reinforced PP layers.

Transverse flow is also an important deformation mode, particularly when forming FMLs with uniaxially oriented UD core at melt-state. During forming, the bent metal covers induce uneven normal pressure onto the FRP core at the bending area, where the polymer could be squeezed outwards transversely. This is also an observed phenomenon when fabricating FRP sheets, where the normal pressure induces a pressure gradient along both of the transverse and longitudinal fibre directions. As a consequence, fibres will flow with the polymer in the transverse direction [140,142], whereas only polymer will be squeezed in the longitudinal direction as the fibres are inextensible [142]. Murtagh and Mallon [140] mentioned that transverse flow in cross-plied UD prepregs or woven fabrics is limited. They also mentioned that the transverse flow can be investigated using parallel-plate flow squeezing apparatus.

6. Defects and challenges in forming FML components

FML forming technologies are under developing with many challenges encountered. There are many common defects, despite their different deformation modes, in forming FMLs and monolithic metal sheets, such as springback, wrinkling, cracking, and thickness variation. Additionally, as a multi-layer material like FRP sheets, delamination can occur in forming FMLs. These defects during forming are the result of the interaction between various factors, including product geometrical design and processing parameters such as temperature, BHF, feed rate, etc. In particular, the stacking arrangement of the layers in FMLs can also affect the forming quality. In this section, typical defects occurred during FML forming and research on their countermeasures are discussed.

6.1. Springback

Springback, as shown in Fig. 14a, is a common defect in sheet metal forming and has been attracting continued research interests [143]. In metal forming, it is caused by elastic recovery which results in geometric inaccuracy, i.e. springback, after forming, and can be difficult to be reduced or eliminated. To improve the geometric accuracy of the products, over-bending the part or increasing local plastic deformation at the radii region is usually considered during the forming process [119]. In FRP forming, however, spring-in occurs because of FRP's anisotropic material properties [144]. Behrens et al. [32] found that the springback effect in metal forming and the spring-in effect of FRP forming occurred at the same time and weakened the counterparty, which would lead to a smaller geometric deviation. Hahn et al. [104] explained that with a certain amount of dwell time, the solidified FRP core can reduce the overall springback when forming FMLs. Mosse et al. [48] reported that Al/PP-FMLs showed 75% less geometric deviation compared to monolithic aluminium when forming a channel. Guo et al. [55] also reported a smaller shape error in forming FMLs compared to monolithic steels. Despite these reported positive results, the multi-material systems also increase the complexity of

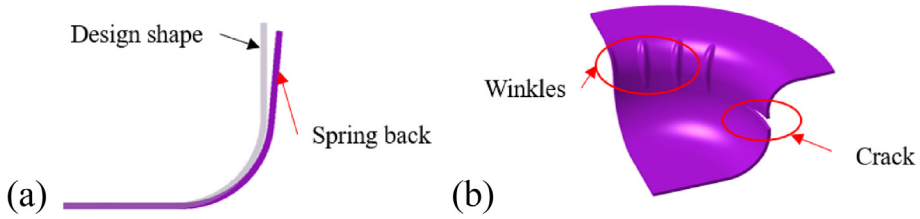


Fig. 14. Typical forming defects: (a) springback, (b) wrinkles and crack.

determining the processing parameters for optimal forming quality as a result of different material processing and deformation mechanisms of metals and FRPs. To obtain parts with better geometric accuracy and less springback when forming FMLs, conventional processing parameters such as the preheating temperature, the BHF and the feed rate need to be reconsidered (as Mosse et al. did in Ref. [47]) and other factors, such as the removal temperature, the cooling rate, the material stack-up, and the die wall angle, were proposed by Behrens et al. [32] for further investigation.

6.2. Wrinkles and cracks

Wrinkling, as shown in Fig. 14b is also a common defect during forming. In sheet metal forming, this is caused by compressive minor stress; and when this stress exceeds a critical value, winkling occurs [145]. When forming laminated materials including FRPs and FMLs, the wrinkling mechanisms are more complicated because of complex loading conditions and deformation modes. For laminates reinforced with woven fibres, the locking of intra-ply shear during forming would cause wrinkling as mentioned in section 5.1 [129,130]. In addition, lack of inter-ply slip would cause wrinkling and cracking in the inner and outer layers of a laminate, respectively [118]. Boisse et al. [146] noted that wrinkling is a global phenomenon and needs to be described with in-plane shear, as well as intra-ply tension and out-of-plane bending. For cross-plied UD laminates, Potter [127] also identified in-plane wrinkling and out-of-plane wrinkling in a bending test.

The conventional solution to reduce wrinkling is to optimise the processing parameters, which has also been investigated in FML forming. For example, BHF, as an important factor in sheet metal forming [147], has also been a major parameter being studied in FML forming. Rajabi and Kadkhodayan [91] indicated that the effect of BHF during FML forming was similar to metal forming, and wrinkling can be eliminated at high BHF provided the laminate tear was being prevented by high temperature. Gresham et al. [83] reported that wrinkling will be significantly more severe with a low BHF at high temperatures. A common conclusion is that higher BHF can reduce wrinkling, but increase the risks of other defects, such as cracks [38].

The blank itself is an important factor for wrinkling control. Rajabi and Kadkhodayan [91] investigated the effect of the initial blank shape with various diameter and thickness values, on drawability. They observed that the blanks with larger diameters would result in a higher drawing force caused by the increasing friction, and the excess material would result in more severe wrinkling. In addition, less severe wrinkling was observed with thicker blanks, which could be explained by the higher buckling resistance resulting from the increase in the thickness.

Furthermore, the multi-material structure of FMLs can aggravate the wrinkling. Mennecart et al. [84] reported that fibre winkles would arise normal forces onto the metal covers, which would cause wrinkling of the covers. They suggested changing the fibre orientation to reduce the fibre wrinkling, which was also a method investigated and validated in preforming dry woven fabrics [129]. Heggemann and Homberg [99] improved the wrinkling condition by forming FMLs comprised steel covers and segmented FRP prepreg patches which were arranged in selective orientations varying from 0° to 135°.

As mentioned above, cracking would occur at a smaller drawing depth with the drawability being limited by excessive BHF. Unlike the cracking occurs in sheet metal forming, where stretch thinning can be observed before the sheet ultimately cracked, not much overall thinning or metal plastic deformation can be observed when FMLs cracks during forming [148] due to the break of inextensible fibres. Wollmann et al. [38] produced a basic drawability diagram, which identified the workable BHF range without cracks or wrinkles corresponding to the drawing depth when forming steel/PA6-based FMLs produced in the LEIKA project.

6.3. Thickness variation

Thickness uniformity has been a critical challenge in forming technologies. In sheet metal forming, most regions of a sheet will experience different degrees of thinning under stretching and ultimately cracking as mentioned above. In FRP forming, limited thinning can be observed before cracking due to the breaking of inextensible fibres under stretching. However, where a pressure gradient is presented, both thinning and thickening can occur due to transverse flow, which would lead to volume fraction variation [149]. The problem of thickness distribution in forming FMLs is usually considered a similar problem as forming FRPs.

Thickness variation in FML forming is normally caused by transverse flow [31]. A higher compaction force at the bottom radii region could cause a reduction in thickness, as illustrated in Fig. 15 [31]. As mentioned in section 5.2, the polymer under unevenly distributed pressure would be squeezed transversely outwards to adjacent regions [55,149], where accumulation would occur and increase the thickness. In addition, polymer accumulation would also occur where the fibre wrinkled, and therefore increase the local thickness [84]. Heggemann and Homberg [99] observed accumulation at the bottom area when drawing a cup. Behrens et al. observed material accumulation at the side wall region during cup-drawing [49] and flange radii when forming a channel [32]. Mennecart et al. [84] also identified similar problem when forming a double-dome component with RTM technique. When the punch was not fully-travelled and sufficient compaction force was not established, the bottom tended to bulge. This bulge disappeared when the punch was fully-travelled and sufficient compaction force was established. Furthermore, polymer extrusion at the flange edge was also reported [31,49].

The direction of fibres affects the thickness variation greatly. Behrens et al. [49] analysed the effects of fibre direction on the polymer extrusion at the flange area when forming FMLs with uniaxially orientated UD prepregs. They explained that the squeezing-out of polymer could not be eliminated, but was less severe along the longitudinal fibre direction than the transverse

fibre direction. This phenomenon can be explained by the transverse flow as discussed in section 5.2. Similar phenomena was observed by Hahn et al. [104] when press brake bending PA6-based FMLs. They noted that the bending area was more likely to thin when the bending line was parallel to the fibre orientation.

As mentioned in section 5.2, the application of woven fabric or cross-plied UD prepregs limits the transverse flow, which can improve the thickness distribution. Another method to improve the thickness distribution is to reduce the pressure gradient. Hahn et al. [104] noted that a larger punch radii can mitigate the thinning phenomenon at the bending region. Furthermore, Heggemann and Homberg [99] filled the glass beads between the metal sheets so that the compaction at the bottom radii was relatively restricted. They also utilised local pre-curing at the bottom area to restrict the polymer flow during stamping.

6.4. Delamination

Delamination is a local interlaminar crack as illustrated in Fig. 15. It is a common failure mode when forming laminated materials, including FRPs and FMLs. Mosse et al. [48] observed delamination at the side wall region when forming fabricated FMLs with an open die. As the open die cannot provide normal surface pressure to prevent the outer metal cover from bending away, they suggested that a closed die would reduce the delamination. They also found that a higher feed rate and temperature could reduce the delamination length. However, excessive temperature resulted in consistently high degree delamination. Dau et al. [31] observed delamination at the flange radii region, and pointed out that inter-ply slip could cause delamination when sufficient surface pressure is absent.

It should be noted that the defects discussed above are normally results of the interactions of several factors instead of one single factor. For example, Rahiminejad et al. [98] reported that as the better inter layer sliding effect is allowed at a higher temperatures, the BHF would have less effect on the severity of the delamination. In the work of DharMalingam et al. [82], a process optimisation using the Taguchi design of experiment was performed to characterise the effects of the preheating temperature, the BHF and the feed rate for process optimisation.

7. Numerical modelling and FEA simulation

Finite element analysis (FEA) is a powerful tool to accelerate the design of the complicated forming processes and reduce the cost. It has been extensively applied in sheet metal forming (reviewed by Ablat and Qattawi [150]) and under rapid development for FRP forming [151,152]. Recently, with the development of FMLs, FEA has also been used to simulate the forming process of FMLs since the late 2000s [33,34,38,51,58,81,82,86,88,89,91–94,97,100]. These research activities were conducted on various commercial FEA packages including Abaqus, LS-DYNA, PAM-Form, etc.

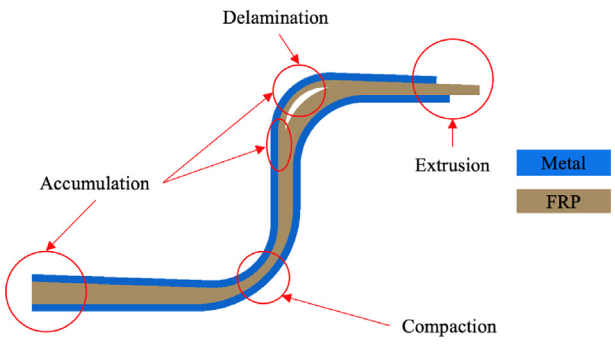


Fig. 15. FMLs forming defects: thickness variation and delamination.

Most of these studies considered FML forming to be a mechanical process at a certain temperature; there are also studies taking account of the thermomechanical properties of materials. Behrens et al. [97] simulated an isothermal forming process of a steel/PA6-based FML at an elevated temperature. Temperature-dependent material behaviour characterised at a range of lower temperatures was used to extrapolate the mechanical properties at the simulated forming temperature. Mosse [94] developed a one way coupled structural-thermal FE model, which enabled the simulation of thermal effects during forming of Al/PP based FMLs. This model was adopted in several subsequent studies: DharMalingam et al. [82] performed an optimisation study by considering processing parameters, including sheet preheating temperature, BHF and feed rate. Dou et al. [88] examined the strain path at representative points at different forming temperatures.

The in-plane deformation of FMLs during forming are normally simulated by treating the metals and FRPs as separate layers. In most cases, the metal covers were modelled with shell elements, adopting isotropic elastic-plastic materials constitutive models. Anisotropic properties of metals were also taken into consideration in some studies [34,51,97]. In particular, Behrens et al. [97] took account of the temperature dependency of the anisotropic coefficients of the steel covers.

The FRP layers were also commonly modelled with shell element; but there have been other approaches developed with various element types and material models. Hahn et al. [58] used membrane elements which feature no bending stiffness to simulate the deformation of UD carbon fibre-reinforced PA6 core. Behrens et al. [97] modelled the UD glass fibre-reinforced PA6 with two layers of shell elements sharing identical nodes. They used an anisotropic hyper-elastic material model and an isotropic elastic-plastic material model for the fibres layer and the polymer matrix layer, respectively. Wollmann et al. [38] used a default Belytchko-Tsay element with elastic stiffness hourglass control in PAM-Form to simulate the woven carbon fibre-reinforced PA6 core. A built-in material model (material 140) was used to model the thermo-viscoelastic-matrix and elastic fibres. Mennecart et al. [33] modelled the injected polymer in the RTM forming with a fluid cavity feature.

The out-of-plane behaviours FMLs during forming have also been simulated in different ways. The bending of metals and FRPs can be captured with the elements and material models selected as discussed above. The interfacial behaviour, especially inter-ply slip, has been drawing particular interest when simulating the FML forming processes. Saadatfard et al. [51] assumed no slippage between the aluminium covers and glass fibre-reinforced polyurethane core, so that the covers and core were tied together in their model. When the interfacial slippage was taken into account, a Coulomb's friction law was usually employed with a coefficient of friction to be defined [34,93]. In addition, cohesive damage behaviour was utilised in some studies to simulate the stick and separation between metal and FRP layers [38,58]. Furthermore, Mosse [94] developed a user-defined interface subroutine to determine whether the bond between aluminium covers and PP-based FRPs was maintained or released for slippage, based on checking against certain time dependent criteria at each time increment of the simulation. Mosse et al. [81] examined the effect of different friction coefficients, which could be a result of different temperatures, in forming FMLs. Compared to the studies on the inter-ply slip, the transverse flow has drawn little attention. Although Behrens et al. [97] mentioned that the anisotropic “intake and squeezing behaviour” of the UD FRP cores can be detected

qualitatively with their "sharing nodes" technique, and Mennecart et al. [33] found a fairly good prediction of the bulging effect at the bottom area with the fluid cavity feature employed, the polymer flow cannot be captured accurately.

There has been a certain degree of progress in simulating FML forming, and FEA has proven to be a useful tool to design, examine and optimise the forming process. With the current modelling techniques as discussed above, most deformation modes during FML forming, including the in-plane behaviour of the constituting layers, out-of-plane bending and inter-ply slip, can be captured. Therefore, FML deformation behaviours as well as various defects, such as wrinkling and excessive thickness reduction (or cracking), can be predicted; and forming quality can thus be evaluated and better analysed by examining certain simulation outputs, such as strain path; however, the transverse flow cannot be accurately captured with the current shell or membrane-based modelling techniques. Therefore, the prediction of the thickness variation caused by polymer flow remains a challenge.

8. Conclusions and future research trends

This review provides an extensive survey and appraisal of the state-of-the-art FMLs and their forming technologies, with key conclusions listed below:

(1) For the material constituents, it can be observed that the materials for metal layers have been extended from aluminium alloys to steels, which allows lower cost thus wider applications in the automotive industry. Thermoplastics have also been replacing thermoset polymers to enable faster production and better recyclability.
(2) Bending of single-curvature stringers and lay-up of large panels with shallow curvatures for FMLs have been relatively well developed and there have been successful applications in the state-of-the-art airplanes. Alternative forming technologies, such as shot peening forming (SPF), incremental sheet forming (ISF), and laser forming, have been analysed and it was evident that they were feasible. However, their limitations in formability and/or productivity make them not the best candidates for the high-volume production of complex-shaped FML components.
(3) Stamp forming has shown promise for producing more complex-shaped FML components, such as deep drawn panels. To achieve better formability, FMLs are usually stamped at elevated temperatures. In addition, other processing parameters and their interactions have also been considered to achieve optimal forming quality. So far, the forming temperature is primarily dependent on the glass transition temperature and melting temperature of the polymer matrix of an FML. The preferable thermomechanical behaviours and forming temperature windows of the metal layers are, however, rarely considered, due to the fact that the most alloys used for producing existing FML structures are of the grades for cold working.
(4) The formability of FMLs is determined by the deformation behaviours of both the metal and FRP layers, as well as their interfacial interactions, during forming. Compared to metals, the understanding of the deformation mechanism of the FRPs and the metal-FRP interfacial behaviours is less adequate; thus, forming defect-free FMLs remains a major challenge. Some defects, including thickness variation caused by transverse flow and delamination, are similar to the defects in FRP forming, which is also drawing significant research attention at the moment. To achieve optimal formability and as-formed quality of FML components, a thorough understanding of their deformation mechanisms under forming conditions is essential.
(5) Modelling and simulation technologies of FML forming processes are being developed alongside the FML forming technologies, and have made certain progress. However, there are gaps that need to be filled. For instance, to accurately capture the transverse flow behaviour is needed for a more comprehensive defect prediction related to polymer squeezing. Therefore, more advanced material constitutive models and inter-layer interfacial models of FMLs are highly desired.

Based on the conclusions drawn above, future research on forming FML components could continue to focus on: (1) Extending the FML material constituents to more extensive material candidates (e.g. new thermoplastic polymers, higher grade alloys, lower-cost steels, etc.), to push the limit of productivity, structural performance, recyclability, and cost reduction of FML components; (2) Making innovations in FML forming processes and forming conditions, by utilising the advanced experience in sheet metal and FRP forming technologies, and optimising the formability of FMLs based on a more in-depth understanding of their deformation mechanisms; (3) Further developing the materials and process modelling based on the understanding of material behaviours during forming to match the new developments in FML forming; and (4) Exploiting the research on new forming technologies to extend the application of FMLs structures to a wider range of vehicles and industrial sectors.

Conflicts of interest

The authors declare that there is no conflicts of interest.

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